3. Scenario

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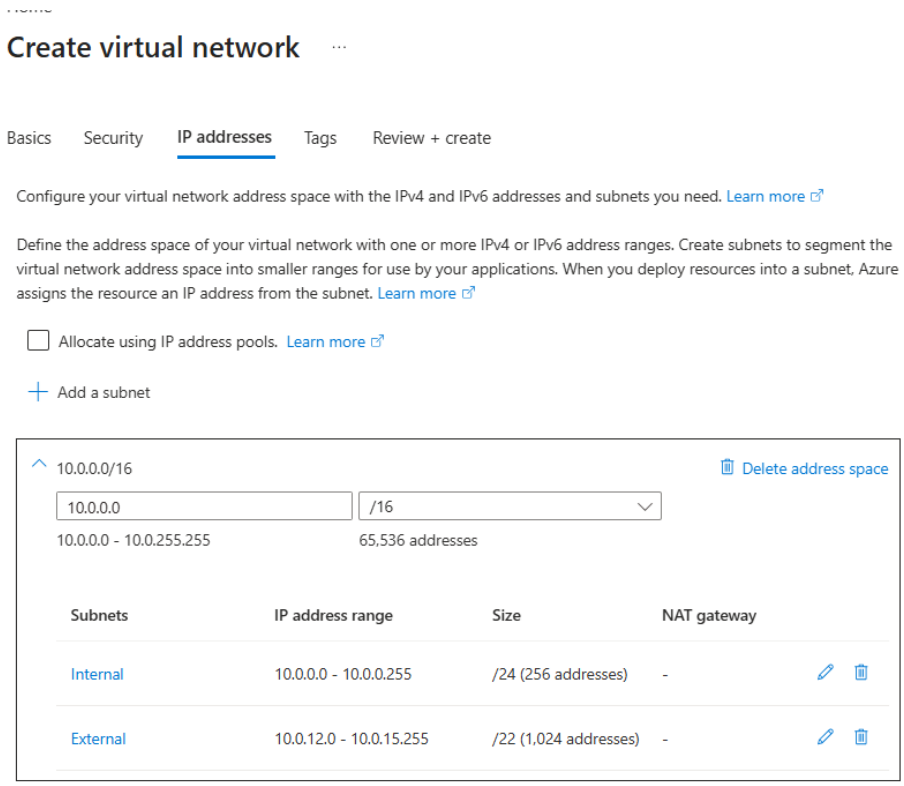
You have been contracted by a small business called **TechBridge Solutions** to design their cloud network infrastructure. The business requires a virtual network that separates public-facing resources from internal back-end systems. Your design must account for at least two distinct network segments.

You are free to determine the specifics of the design — address ranges, number of subnets, and security rules — but your choices must be documented and justified.

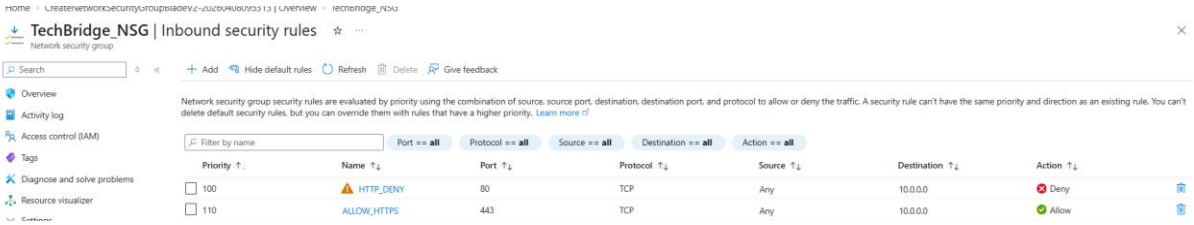
4. Tasks and Mark Allocation

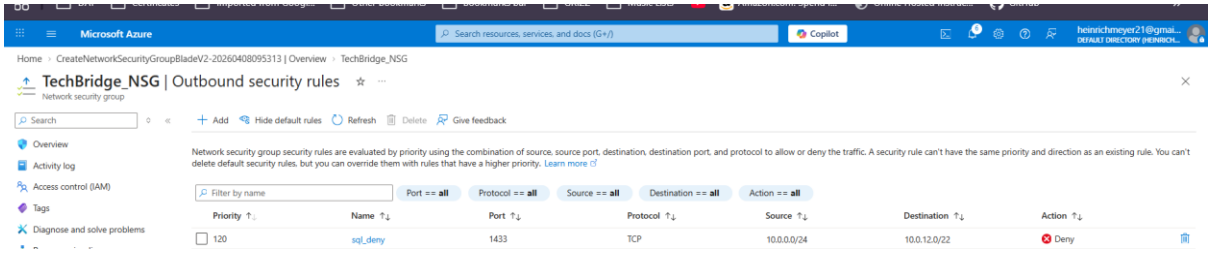
Task 1 and 2:



I've selected the Class A private address as it contains many addresses that can be taken advantage of if the network expands, I've also created 2 subnets one for internal back-end systems and one for External public facing resources. I've also ensures that the External has more usable hosts to accommodate the additional traffic it needs to be able to handle.

Task 3





There are 3 NSG rules 2 inbound and 1 outbound, Although this is a rather basic NSG config it should cover the basics for **TechBridge.** The 2 inbound rules are specially for regulating the traffic that is allowed to access the 10.0.0.0 network. It prevents network from HTTP as it is not a secure connection and inversely it allows HTTPS because it is secure. The singular outbound rule is to prevent SQL traffic from moving from the Internal subnet to the External subnet. This is just to ensure that the companies data is kept more secure.